

HOMEWORK - CHAPTER 2, PART 2 & 3

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Q9. Which one of the four representations for signed integers is used most often by digital computer systems? (Signed magnitude, one's complement, two's complement, excess-m)

Two's complement is the most common method of representing signed integers on computers

Q15. What are the goals of Booth's algorithm?

One of the goals for Booth's algorithm is to create an effective binary multiplication algorithm that speeds up huge numbers with fewer addition operations. Booth's algorithm also makes good use of two's complement numbers and symmetric bit patterns.

Q. 72 When would you choose a CRC code over a Hamming code? A Hamming code over a CRC?

CRC codes are good when efficient error detection without error correction is required, but hamming codes are appropriate when both error detection and correction are required, particularly for single-bit errors.

E16. Represent the following decimal numbers in binary using 8-bit signed magnitude, one's complement, two's complement, and excess-127 representation:

1. 78 = Ones- 10110001 Twos-10110010 Signed- 01001110 Excess- 11001101
2. -44= Ones- 00101011 Twos-00101100 Signed- 11010100 Excess - 01010011
3. 118 = Ones- 10001001 Twos-10001010 Signed- 01110110 Excess- 11110101
4. -117 = Ones- 01110100 Twos- 01110101 Signed- 1000101 Excess- 00001010

E34. Subtract the following signed binary numbers as shown using 2's complement arithmetic.

1. $01110101 - 00111011 = 117 - 59 = 58$

2. $00110101 - 00001011 = 53 - 11 = 42$

3. $01101111 - 00010001 = 111 - 17 = 94$

E36. Perform the following binary multiplications, assuming unsigned integers:

1. $1100 \times 101 = 00111100 = 60$

2. $10101 \times 111 = 10010011 = 147$

3. $11010 \times 1100 = 00111000 = 312$

E38. Perform the following binary divisions, assuming unsigned integers:

1. $101101 \div 101 = 000010010 = 18$

2. $10000001 \div 101 = 00011001 = 25$

3. $1001010010 \div 1011 = 00110110 = 54$

E45. If the floating-point number representation on a certain system has a sign bit, a 3-bit exponent and a 4-bit significand:

1. What is the largest positive and the smallest positive number that can be stored on this system if the storage is normalized? (Assume no bits are implied, there is no biasing, exponents use two's complement notation, and exponents of all zeros and all ones are allowed.)

The biggest positive number in this format, if the value was composed of a 3-bit exponent and a 4-bit mantissa, would be:

0 111 1111

Which equates to 15.5.

2. What bias should be used in the exponent if we prefer all exponents to be non-negative? Why would you choose this bias?

You can select a bias of three if the exponent has three bits. All exponents can be displayed in a non-negative manner with a bias of three.